

SRM VALLIAMMAI ENGINEERING COLLEGE

(An Autonomous Institution)

SRM Nagar, Kattankulathur, Chengalpattu Dt.-603203, Tamil Nadu.

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

ACADEMIC YEAR: 2024-2025

REPORT

on

HEARTHIVE

MINI PROJECT EXPO



SUBMITTED BY

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THIRD SEMESTER

2024-2025 (Odd Semester)

BATCH: 2023-2027

VISION

To devise captivating, fascinating and unique practices of teaching that discovers the trained talent and inherent competences of young minds to evolve as humane professional Computer Science Engineers.

MISSION

- To provide students with challenging ventures, contributing to the betterment of their selfdom to compete with international talents.
- To act as a motivational hub to exhibit practical knowledge with the latest technological updates and research publications.
- To render ample knowledge to exhibit their ubiquitous talents for the social prosperity and promote industry-institute harmony to upgrade the standards for the international reputation.

PROGRAMME EDUCATIONAL OBJECTIVES (PEO's):

- **PEO1:** To mould students to exhibit top performance in the higher education and research and to become the State-of-the-art technocrat.
- **PEO2:** To impart the necessary background in Computer Science and Engineering by providing solid foundation in Mathematical, Science and Engineering fundamentals.
- **PEO3:** To equip the students with the breadth of Computer Science and Engineering innovate novel solutions for the benefit of common man.
- **PEO4:** To groom the students to be multifaceted entrepreneurs with professional ethical attitude in broader social perspective.
- **PEO5:** To provide an ambience learning environment that is conducive for the growth of successful professional career of students.

PROGRAM SPECIFIC OUTCOMES (PSO's):

PSO1: Exhibit proficiency in planning, implementing and evaluating team oriented-software Programming solutions to specific business problems and society needs.

PSO2: Demonstrate professional skills in applying programming skills, competency and decision making capability through hands-on experiences.

PSO3: Apply logical thinking in analyzing complex real world problems, and use professional and ethical behaviors to provide proper solutions to those problems.

PSO4: Demonstrate the ability to work effectively as part of a team in applying technology to business and personal situations

PROGRAMME OUTCOMES (PO'S) :

Engineering Graduates will be able to

1. Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using the first principles of mathematics, natural sciences, and engineering sciences.
3. Design/development of solutions: Design solutions for the complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. Conduct investigations of complex problems: Use research-based knowledge and research methods including the design of experiments, analysis and interpretation of data, and the synthesis of the information to provide valid conclusions.
5. Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

6. The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12. Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

OBJECTIVE

The primary objective of this project is to design and develop a sophisticated Medical Retrieval-Augmented Generation (RAG) chatbot utilizing the BioMistral open-source Large Language Model (LLM). This chatbot will serve as an advanced tool for delivering accurate, contextually relevant, and reliable responses to user queries specifically focused on heart health. By integrating BioMistral LLM with a comprehensive heart health knowledge base stored in PDF format, the system will enable the retrieval of relevant, factual medical information to generate well-rounded and informative answers.

The project aims to create a platform that not only understands user queries but also draws on documented and validated medical resources to provide detailed, context-driven responses. This will enhance the accuracy and trustworthiness of the information, particularly in addressing key topics such as cardiovascular diseases, prevention strategies, treatments, and lifestyle recommendations. The chatbot will be designed to cater to a wide audience, from patients and caregivers to healthcare professionals, offering an accessible, user-friendly interface for gaining insights into heart health.

Through this project, the ultimate goal is to reduce the information gap in the field of cardiovascular health by creating an interactive tool that combines state-of-the-art AI technology with expert-validated knowledge. This chatbot will support better-informed decisions, promote heart health awareness, and empower users to take proactive steps in managing and improving their cardiovascular well-being. By making reliable medical information more accessible, the project seeks to contribute meaningfully to public health and healthcare education, fostering a deeper understanding of heart health issues.

INTRODUCTION

In recent years, the integration of Artificial Intelligence (AI) into the healthcare industry has led to profound changes in the way medical services are delivered and consumed. With the ever-increasing demand for accessible, accurate, and personalized healthcare, AI-powered systems are becoming essential in bridging the gap between patients and healthcare providers. One of the most promising advancements in this area is the development of intelligent chatbots capable of answering medical queries in real-time, offering users a reliable source of information at their fingertips.

This project aims to develop a Medical Chatbot using the BioMistral Large Language Model (LLM), a state-of-the-art open-source model, optimized for processing and generating responses to medical language. By leveraging a heart health-related PDF document as the primary knowledge base, the chatbot is designed to provide users with accurate, contextually relevant responses to their queries on cardiovascular health. The chatbot employs a powerful method known as Retrieval-Augmented Generation (RAG), which enhances the accuracy of its responses by first retrieving relevant information from the heart health document and then generating a response based on the retrieved data.

The Need for AI-Powered Healthcare Solutions

The global healthcare system is facing immense challenges, particularly with increasing populations, a growing prevalence of chronic diseases, and the need for more personalized care. These challenges have driven the demand for AI-powered healthcare solutions, which can provide real-time, accurate, and accessible medical information. Chatbots, in particular, have emerged as a key technology that can reduce the burden on healthcare professionals while offering immediate assistance to patients.

A medical chatbot powered by AI can serve various purposes, including providing initial assessments, offering guidance on treatment options, and delivering health education. This chatbot, focused on heart health, addresses a critical area where patients require continuous support and information. Cardiovascular diseases (CVDs) are the leading cause of death globally, accounting for nearly 17.9 million deaths annually. Despite the high prevalence of heart conditions, many individuals still face barriers in accessing timely and reliable medical advice, whether due to geographical limitations, financial constraints, or limited access to healthcare professionals.

AI-powered chatbots can address these issues by offering 24/7 assistance, delivering medically accurate information, and enhancing patient engagement with their own health. For instance, the chatbot developed in this project allows users to ask questions related to heart disease prevention, diagnosis, symptoms, treatments, and healthy lifestyle practices. By doing so, it provides users with crucial information that can help them make informed decisions about their health.

BioMistral LLM: Optimized for Medical Language

At the core of this chatbot is the BioMistral LLM, an open-source language model specifically optimized for understanding and generating medical language. Medical language, especially in fields like cardiology, can be highly specialized, with complex terminologies and concepts that are often difficult for general language models to process accurately. BioMistral has been trained on vast amounts of medical data, allowing it to understand these terminologies and provide responses that are contextually accurate and medically sound.

This is particularly important in healthcare applications, where the quality of information can significantly impact patient outcomes. A chatbot providing inaccurate or incomplete medical advice could lead to confusion or poor health decisions.

The Role of Retrieval-Augmented Generation (RAG)

While the BioMistral LLM is a powerful model capable of generating accurate responses, the Retrieval-Augmented Generation (RAG) approach enhances its performance by ensuring that the information provided is always grounded in factual, documented sources. The RAG method combines two key processes: retrieval and generation. When a user submits a query, the system first retrieves relevant sections of the heart health PDF document, ensuring that the response is based on authoritative information. Once the information is retrieved, the model then generates a response that is both accurate and easy to understand.

This two-step process ensures that users receive fact-based answers that are contextually appropriate to their questions. For example, if a user asks about the symptoms of heart disease, the RAG system will first locate the relevant sections in the heart health document before generating a detailed and user-friendly explanation. By combining retrieval and generation, the chatbot offers a higher degree of reliability compared to traditional chatbots that rely solely on generation models.

Heart Health: A Critical Global Concern

The focus of this project on heart health is both timely and relevant. Cardiovascular diseases are the leading cause of death globally, and they affect individuals across all demographics. From heart attacks and strokes to chronic conditions like hypertension and coronary artery disease, heart health is a major concern for millions of people worldwide. Early detection, prevention, and proper management of heart conditions are essential for reducing the global burden of these diseases.

However, many people lack access to the necessary resources to educate themselves about heart health. Whether it is due to financial constraints, lack of medical expertise, or limited access to healthcare services, there are significant barriers that prevent individuals from gaining the information they need.

This project addresses this gap by providing a reliable, AI-powered solution that delivers essential heart health information in an accessible format. Users can interact with the chatbot to ask questions about heart disease prevention, identify symptoms of potential heart issues, or seek advice on healthy lifestyle changes that promote better heart health.

Impact of the Project

This project is designed to have a significant impact on the way people access and understand heart health information. By using an AI-powered chatbot, users can gain immediate access to reliable medical advice, reducing their dependence on healthcare providers for initial consultations. This is particularly important for individuals living in remote areas or for those who may not have access to regular medical care.

The chatbot also promotes patient education and engagement, empowering users to take a more active role in managing their heart health. By offering personalized, accurate information, the chatbot can help users make informed decisions about their lifestyle, health habits, and when to seek medical attention. This could potentially lead to better health outcomes, especially for those at risk of developing cardiovascular diseases.

Moreover, the integration of the BioMistral LLM and the RAG framework showcases the potential of AI in healthcare applications. As more AI-driven solutions are developed, the ability to offer specialized, reliable, and user-friendly tools for patient education and support will continue to grow. This project serves as a demonstration of how AI technology can be leveraged to address critical health concerns, improving access to medical information and contributing to better global health outcomes.

PROJECT DESCRIPTION

The primary goal of this project is to develop a sophisticated Medical Chatbot leveraging the power of the BioMistral Large Language Model (LLM) and using a heart health-related PDF document as the core knowledge base. This chatbot is designed to provide accurate, context-aware responses to queries about heart health by employing an advanced technique known as Retrieval-Augmented Generation (RAG).

The BioMistral LLM is an open-source language model that is particularly well-suited for handling medical text and specialized terminologies. Medical fields, especially cardiology, involve the use of complex and highly specific language, making it necessary to use a model like BioMistral, which has been optimized for processing this type of information. The LLM allows the chatbot to manage a broad spectrum of inquiries related to heart health, including prevention measures, potential treatments, symptoms of heart diseases, diagnosis guidance, and general heart health advice.

The chatbot accesses a heart health PDF document as its primary knowledge source, ensuring that the responses are grounded in credible, documented medical information. By combining this robust knowledge base with the capabilities of BioMistral, the chatbot can ensure the accuracy and reliability of the information it provides to users.

Key Project Components:

Data Integration: The heart health-related PDF serves as the central knowledge base from which the chatbot draws information. The PDF is integrated into the system so that the chatbot can retrieve relevant information from the document based on user queries. This ensures that the chatbot's responses are backed by authoritative medical content, making the chatbot both a reliable and efficient tool for users seeking heart health advice.

Model Implementation: The project utilizes the BioMistral LLM, which is specifically designed for medical language understanding. This model interprets and processes the user's natural language queries, ensuring that even complex and specialized medical questions are understood accurately. The model uses sophisticated language understanding techniques to generate medically sound responses by drawing from the knowledge stored in the PDF.

RAG (Retrieval-Augmented Generation) Framework: The Retrieval-Augmented Generation technique plays a critical role in enhancing the chatbot's performance. The RAG model first retrieves the most relevant data from the heart health PDF based on the user's input. After retrieving the appropriate sections, it generates a contextually appropriate response. This two-step process ensures that the chatbot delivers highly relevant, fact-based, and personalized answers to user questions. The RAG framework combines both retrieval-based techniques, which ensure accuracy, and generative techniques, which allow for natural and conversational responses.

User Interaction: The chatbot provides an intuitive and interactive experience for users by allowing them to ask questions in natural language. Users do not need to have any technical or medical expertise to use the chatbot. The system is designed to handle questions about various aspects of heart health and provide clear, concise answers in a conversational format. This aspect of user interaction ensures that the chatbot is accessible to a broad audience, including patients, caregivers, or anyone interested in learning more about heart health.

Motivation and Impact:

The healthcare industry is increasingly embracing AI-driven solutions to make medical information more accessible. There is a growing need for medical chatbots that can provide real-time assistance to individuals seeking reliable health information. Heart health is one of the most crucial areas of healthcare, as cardiovascular diseases remain one of the leading causes of death globally. Many people require timely access to trusted medical advice regarding heart disease prevention, symptoms, and treatment, but often face barriers to accessing healthcare professionals.

This project addresses the need for accurate, trustworthy, and easily accessible heart health information by developing a user-friendly chatbot that can provide real-time assistance. The chatbot aims to:

Improve Accessibility: By making heart health information available in a conversational format, the chatbot lowers the barrier to medical advice, making it more accessible to people who may not have immediate access to healthcare professionals.

Reduce Information Gaps: The chatbot provides a bridge between users and the complex medical literature, helping users understand detailed medical information in an easily digestible manner.

Enhance User Experience: With a focus on natural language understanding, the chatbot offers a smooth, interactive experience, allowing users to ask questions in a conversational style and receive clear, informative answers.

CODE

Hearthive

1 Q (x) >- Build BioMistral Medical RAG Chatbot using BioMistral Open Source LLM

```
[ ]: from google.colab import drive
drive.mount('/content/drive')
```

Medical Chatbot with BioMistral LLM and Heart Health pdf file.

1.1 Load the google drive

```
[ ]: from google.colab import drive
drive.mount("/content/drive")
```

Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).

1.2 Installation

```
[ ]: !pip install langchain sentence-transformers chromadb llama-cpp-python
↳ langchain_community pypdf
```

1.3 Importing libraries

```
[ ]: from langchain_community.document_loaders import PyPDFDirectoryLoader
from langchain.text_splitter import RecursiveCharacterTextSplitter
from langchain_community.embeddings import SentenceTransformerEmbeddings
from langchain.vectorstores import Chroma
from langchain_community.llms import LlamaCpp
from langchain.chains import RetrievalQA, LLMChain
```

1.4 Import the document

```
[ ]: loader=PyPDFDirectoryLoader("/content/drive/MyDrive/mypro/data")
docs=loader.load()
```

```
[ ]: len(docs) # number of pages
```

```
[ ]: docs[6]
```

1.5 Chunking

```
[ ]: text_splitter=RecursiveCharacterTextSplitter(chunk_size=300,chunk_overlap=50)
chunks=text_splitter.split_documents(docs)
```

```
[ ]: len(chunks)
```

```
[ ]: chunks[600]
```

```
[ ]: chunks[601]
```

1.6 Embeddings creations

```
[ ]: import os
os.environ['HUGGINGFACEHUB_API_TOKEN']="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"
```

```
[ ]: embeddings = SentenceTransformerEmbeddings(model_name="NeuML/
↳pubmedbert-base-embeddings")
```

1.7 Vector Store creation

```
[ ]: vectorstore = Chroma.from_documents(chunks,embeddings)
```

```
[ ]: query = "who is at risk of heart disease?"

search_results = vectorstore.similarity_search(query)
```

```
[ ]: search_results
```

```
[ ]: retriever = vectorstore.as_retriever(search_kwargs={'k':5})
```

```
[ ]: retriever.get_relevant_documents(query)
```

1.8 LLM Model loading

```
[ ]: llm=LlamaCpp(
    model_path="/content/drive/MyDrive/mypro/BioMistral-7B.Q4_K_M.gguf",
    temperature=0.2,
    max_tokens= 2048,
    top_p=1
)
```

1.9 Use LLM and retriever and query , to generate final response

```
[ ]: template = """
<|context|>
You are an Medical Assistant that follows the instructions and generate the
    accurate response based on the query and the context provided.
Please be truthful and give direct answers.
</s>
<|user|>
{query}
</s>
<|assistant|>
"""
```

```
[ ]: from langchain.schema.runnable import RunnablePassthrough
from langchain.schema.output_parser import StrOutputParser
from langchain.prompts import ChatPromptTemplate
```

```
[ ]: prompt = ChatPromptTemplate.from_template(template)
```

```
[ ]: rag_chain = (
    {"context":retriever, "query": RunnablePassthrough()}
    | prompt
    | llm
    | StrOutputParser()
)
```

```
[ ]: response = rag_chain.invoke(query)
```

```
[ ]: response
```

```
[ ]: import sys

while True:
    user_input = input(f"Input query: ")
    if user_input == 'exit':
        print("Exiting...")
        sys.exit()
    if user_input == "":
        continue
    result = rag_chain.invoke(user_input)
    print("Answer: ",result)
```

OUTPUT

Input query: where is heart located?

Llama.generate: 54 prefix-match hit, remaining 16 prompt tokens to eval

llama_perf_context_print: load time = 38001.14 ms

llama_perf_context_print: prompt eval time = 0.00 ms / 16 tokens (0.00 ms per token, inf tokens per second)

llama_perf_context_print: eval time = 0.00 ms / 70 runs (0.00 ms per token, inf tokens per second)

llama_perf_context_print: total time = 70090.13 ms / 86 tokens

Answer: The heart is a muscular organ that pumps blood through the circulatory system. It is located in the chest, just behind and slightly left of the breastbone (sternum). The size of the heart varies depending on the person's age, sex, and overall health. In general, it is about the size of a fist.

Input query: What Is Atherosclerosis?

Llama.generate: 54 prefix-match hit, remaining 20 prompt tokens to eval

llama_perf_context_print: load time = 38001.14 ms

llama_perf_context_print: prompt eval time = 0.00 ms / 20 tokens (0.00 ms per token, inf tokens per second)

llama_perf_context_print: eval time = 0.00 ms / 74 runs (0.00 ms per token, inf tokens per second)

llama_perf_context_print: total time = 71016.01 ms / 94 tokens

Answer: Atherosclerosis is a condition in which plaque builds up inside the arteries. This can reduce blood flow and may lead to heart attack, stroke or other health problems. It is caused by many factors including smoking, high cholesterol, high blood pressure, diabetes, obesity, lack of exercise, ageing and family history.

Input query: exit

Exiting...